

# Time-blind by Design: Diagnosing Temporal Traceability in Transformer Representations

Jefferson O. Silva<sup>1,2</sup>

<sup>1</sup> Instituto de Tecnologia e Liderança, São Paulo, Brazil  
jefferson.oliveira@prof.inteli.edu.br

<sup>2</sup> Pontifícia Universidade Católica de São Paulo, São Paulo, Brazil

**Abstract.** Words do not preserve fixed meanings across time. Without suitable representations, we cannot directly compare uses of *gay* in 1920 and 1980. Yet time-conditioned transformers have rarely been tested on whether the neighborhoods of their contextual representations reflect how a word is used in the requested period. Recovering the period identifier alone does not ensure this property, which we call *temporal traceability*. We introduce a controlled test suite with predefined stable, drifting, and bifurcating usage trajectories to evaluate it.

Three findings emerge. Giving the model period information more than doubles the change in neighborhood composition for drifting subjects relative to the non-temporal baseline. However, direct addition of the period vector and a learned token-period projection produce equivalent average changes when nearby verbs and objects are reliable cues; a difference appears when those cues are degraded. Finally, memory ablations show that label-derived history improves later-period classification accuracy, whereas mean pooling and two  $k$ -means centers do not surpass Token-Time.

**Keywords:** Temporal semantics · Semantic change · Transformer models · Traceability · Diagnostic evaluation

## 1 Introduction

Words do not preserve fixed meanings across time. In 1900, *broadcast* referred to scattering seeds across a field; today it describes a social media post reaching millions. Lexical semantic change detection (LSCD) usually learns vector representations from word co-occurrence in a separate corpus for each period and then compares them [10, 11]. This approach has produced important methods and benchmarks, but it summarizes change as one similarity score between separately trained models: **broadcast@1900** and **broadcast@2020** come from different models and become comparable only after their spaces are aligned, rather than being explicit queries to one model.

We call contextual representations *temporally traceable* when representations of the same word conditioned on requested periods are directly comparable and changes in their nearest-neighbor composition reflect changes in how the word

is used. Consider *gay*, which in the early twentieth century was mainly associated with *cheerful* and *merry*, and by the late twentieth century with *queer* and *lesbian* [5]. Comparable representations can support socially relevant questions, including whether a word moved toward or away from stigmatized terms. Answering such questions requires directly comparable representations of uses of *gay* conditioned on 1920 and 1980.

In this study, we test temporal traceability in a controlled environment where the usage trajectory is predefined. In natural corpora, an apparent change may instead reflect differences in word frequency, genre, or corpus composition. It is therefore difficult to tell whether a change in the learned representations comes from the model architecture or the data. We introduce a corpus in which subjects follow predefined stable, drifting, or bifurcating usage trajectories. Four diagnostics distinguish changes in neighborhood composition, recovery of the generating group, sensitivity to the period, and transfer to later periods.

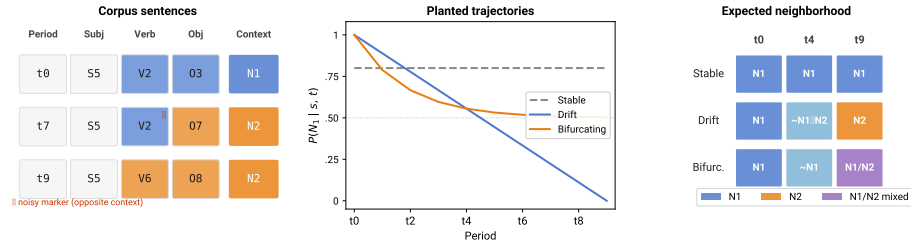
Against this suite we evaluate one non-temporal baseline (Standard) and three time-aware architectures. Additive adds the same period vector at every position. Token-Time instead projects each token–period pair, and Memory-Augmented also summarizes earlier periods. We ask whether these designs improve temporal traceability and what each contributes.

The paper makes four contributions. (1) We define temporal traceability through changes in the neighborhoods of comparable, period-conditioned contextual representations. (2) We introduce a controlled corpus with three usage trajectories and four diagnostics that separate neighborhood composition from label prediction. (3) We show that all time-aware architectures more than double the change for drifting subjects; Additive and Token-Time are equivalent with reliable cues, while Token-Time performs better when they are uninformative. (4) Memory ablations show that label-derived history improves later-period classification, whereas representation-derived means and two  $k$ -means centers do not surpass Token-Time.

## 2 Related Work

Lexical semantic change detection (LSCD) is the dominant computational framing for meaning shift. Diachronic word embeddings [5] learn one vector space per time slice, align the separately trained spaces, and measure how a word’s neighbors change. We share the view that meaning change should alter these neighbors, but use a different architecture: LSCD produces a score comparing two spaces rather than period-conditioned contextual representations from one model. Standard benchmarks formalize this task [10, 11], but changes in frequency and genre can also affect their scores; Spanish shared tasks show that manual annotation is required even for modern corpora [13]. Our controlled corpus makes each usage trajectory known, allowing differences to be attributed to model design rather than corpus composition.

Temporal neural language models avoid separate spaces by conditioning one model on time. Prior work uses timestamps for knowledge-intensive QA, masked



**Fig. 1.** Controlled diagnostic suite overview. *Left:* SVO sentences for a drifting subject across three periods; an  $N_1$  verb appearing in a sentence generated from  $N_2$  at  $t7$  (marked \*) illustrates the deliberate noise in contextual cues. *Center:* predefined  $P(N_1 | s, t)$  trajectories for Stable (flat), Drift (monotonic), and Bifurcating (levelling at 0.5) subjects across  $t0$ – $t9$ . *Right:* nearest-neighbor behavior expected from the predefined trajectory.

language modeling, or period-stratified downstream evaluation [3, 9, 8]. These results show that providing time can improve predictions for the requested period, but do not test whether the neighborhoods of period-conditioned word representations can be inspected and compared directly.

Finally, semantic change may involve bifurcation rather than monotonic drift. Giulianelli et al. show that words can develop coexisting senses [4]; a mean-pooled summary can blur their distinction by falling between them. Our stable, drifting, and bifurcating trajectories test whether one mean summary per subject and period is adequate, and the memory ablations examine alternatives.

### 3 Controlled Diagnostic Suite

We construct a corpus of subject–verb–object (SVO) sentences spanning ten periods. Every sentence consists of one subject, one verb, and one object. We divide the 8 verbs and 8 objects into two groups,  $N_1$  and  $N_2$ , which provide distinct contextual cues:  $N_1$  uses  $\{V1, \dots, V4\}$  and  $\{O1, \dots, O4\}$ ;  $N_2$  uses  $\{V5, \dots, V8\}$  and  $\{O5, \dots, O8\}$ . A sentence generated from  $N_1$  therefore looks like S5 V2 O3, while one from  $N_2$  looks like S5 V6 O8; the subject token is the same in both cases, but its co-occurring words differ. For each of the 30 subjects,  $P(N_1 | s, t)$  is the probability that a sentence containing subject  $s$  at period  $t$  is drawn from  $N_1$ ; its evolution from  $t0$  through  $t9$  defines the subject’s predefined trajectory.

Figure 1 (center) shows the three predefined trajectory classes. Stable subjects maintain an approximately constant  $P(N_1 | s, t)$  throughout. Drift subjects follow a monotonic trajectory from near-exclusive  $N_1$  ( $P(N_1 | s, t) \approx 1$  at  $t0$ ) to near-exclusive  $N_2$  ( $P(N_1 | s, t) \approx 0$  at  $t9$ ): a drifting subject that appears in sentences like S5 V2 O3 early on will appear in sentences like S5 V6 O8 by the final period. Bifurcating subjects begin in  $N_1$  and move toward a mixed state ( $P(N_1 | s, t) \approx 0.5$ ) where sentences are generated from both groups with equal

probability, approximating two coexisting senses. The suite contains 10 subjects per class.

Each verb and object is drawn from the group opposite to the sentence’s generating group 25% of the time (Figure 1, left). The accompanying words are therefore informative but imperfect cues; subject identity and period provide the prior  $P(N_1 | s, t)$ , while the generating group of each sentence remains stochastic. The `true_context` field records this generating group. It is not used in MLM training and is available only to the diagnostics.

This design separates differences caused by the architecture from differences in frequency, genre, or corpus composition. Because  $P(N_1 | s, t)$  is known, we can test whether the composition of representation neighborhoods follows the process that generated the corpus. The scope of these controlled diagnostics is revisited in the Discussion.

The experiments use four evaluation settings. The standard and ambiguous cue conditions vary the reliability of the accompanying words. In the standard cue condition, after  $N_1$  or  $N_2$  is selected to generate a sentence, each verb and object is drawn from that same group with 75% probability, matching the distribution used for masked language modeling (MLM) training. In the ambiguous cue condition, this probability is reduced to 50%, so the verb and object do not reveal which group generated the sentence on average. The contrastive set presents the same masked-verb subject–object frame with two period identifiers and measures whether changing only the period changes the preferred generating group. The later-period transfer set contains periods  $t8$ – $t9$ , which are excluded from MLM training on  $t0$ – $t7$ .

## 4 Time-Aware Architectures

We evaluate one non-temporal baseline and three time-aware architectures, all using the same Transformer encoder configuration. Standard (no period input) is the non-temporal baseline. The three time-aware architectures differ in how they introduce time: Additive gives every token the same period vector; Token-Time uses a learned joint token–period projection before the encoder; and Memory-Augmented also lets the subject consult summaries of earlier periods.

All models follow the same pipeline based on the Transformer encoder [12]. A sentence is tokenized as `[CLS] S V O [SEP]`. The learned token embedding  $e(w_i)$  is combined with position and, when available, period information to form the encoder input  $z_i \in \mathbb{R}^d$ . The sequence  $(z_1, \dots, z_n)$  is processed by a Transformer encoder, and the final hidden state at the subject position,

$$(h_1, \dots, h_n) = \text{Encoder}(z_1, \dots, z_n), \quad h_s = h_{\text{subj}},$$

is the contextual representation of that subject occurrence inspected by the diagnostics. All models are trained with MLM [2]: some tokens are masked, and the model reconstructs them from the remaining tokens and, when available, the period input. The variants differ in how they inject temporal information before the encoder.



**Fig. 2.** Architecture comparison. **Standard**: no temporal signal. **Additive**: adds the period vector to every token. **Token-Time**: uses a joint token–period projection. **Memory-Augmented**: adds past-period summaries and attention over them. New components are yellow; dashed grey boxes are inactive.

The three time-aware models use the same type of period encoding. Within each model,  $\tau(t) \in \mathbb{R}^d$  is a learned projection of sinusoidal features of  $t/(T-1)$ , where  $T = 10$  is the number of corpus periods. It is distinct from the learned position embedding  $p_i$ , which marks each token’s position. Additive and Token-Time differ in how they combine  $\tau(t)$  with token identity; Memory-Augmented lets the subject input consult summaries from earlier periods. Figure 2 summarizes the comparison.

#### 4.1 Baseline: Standard Transformer

Standard is the control architecture: no period signal is available, so any temporal behavior must arise from changes in the observed words alone.

$$z_i = e(w_i) + p_i.$$

#### 4.2 Additive Time-Conditioned Transformer

Additive tests the simplest use of time: adding the same period vector  $\tau(t)$  to every token:

$$z_i = e(w_i) + p_i + \tau(t).$$

S5 V2 03 therefore receives different inputs at  $t0$  and  $t9$ : each token is combined with  $\tau(0)$  or  $\tau(9)$ . The same period vector is added at every position, so time enters each token in the same way.

### 4.3 Token-Time Transformer

Token-Time tests a learned alternative to direct addition. It concatenates each token embedding with  $\tau(t)$  and projects the resulting  $2d$ -dimensional vector back to  $d$  dimensions:

$$z_i = W \cdot \text{concat}(e(w_i), \tau(t)) + p_i,$$

where  $\text{concat}$  joins the two  $d$ -dimensional vectors into a  $2d$ -dimensional vector, and the learned projection matrix  $W \in \mathbb{R}^{d \times 2d}$  maps it back to  $d$  dimensions. The comparison with Additive tests whether this projection helps beyond adding the period vector directly.

### 4.4 Memory-Augmented Extension

The Memory-Augmented extension tests whether representations from earlier periods add useful information beyond the current token-period input. It begins with the same encoder inputs as Token-Time. For a sentence containing S5 at  $t7$ , it builds encoder inputs for [CLS] S5 V 0 [SEP] and modifies only the subject input before the Transformer encoder sees the sentence.

The memory stores one past-period mean summary  $m(s, t)$  for each subject and period. This vector is the mean of the final hidden states  $h_s$  from MLM training sentences containing subject  $s$  at period  $t$ . When processing a sentence at period  $t$ , the model can access only summary vectors from earlier periods. At  $t7$ , for example, it can use the S5 vectors from  $t0$  through  $t6$ , but not from  $t7$  or later.

The model uses attention to combine these past vectors. The current subject input  $z_s$  is the attention query, while  $M_s$ , with one past-period vector per row, provides the keys and values:

$$r_s = \text{Attn}(z_s, M_s, M_s).$$

The resulting vector  $r_s$  is a weighted combination of the past-period vectors. A learned scalar gate  $g$ , bounded by  $\tanh$ , controls how much this summary changes the current subject input:

$$\tilde{z}_s = z_s + \tanh(g) \cdot (\text{LN}(z_s + r_s) - z_s).$$

Here, LN denotes layer normalization. Because  $g$  is initialized at zero, the extension initially behaves exactly like Token-Time. Finally,  $\tilde{z}_s$  replaces  $z_s$  in the input sequence. Verb and object inputs are unchanged, so the subject is the only token that enters the encoder with information from earlier periods.

## 5 Evaluation Protocol and Diagnostics

We evaluate  $h_s$  with four diagnostics. Every time-aware architecture is compared with Standard. We also compare Token-Time with Additive and Memory-Augmented with Token-Time. These contrasts measure the effects of providing time, using a learned projection instead of direct addition, and consulting earlier representations.

**D1: Linear classifier [1].** D1 tests whether a linear classifier can tell from  $h_s$  whether the sentence was generated from  $N_1$  or  $N_2$ . We freeze the encoder and train the classifier on representations from the standard cue condition. Its accuracy on these same representations is training accuracy; the ambiguous cue condition tests it without retraining. High accuracy does not show how the representation’s nearest neighbors change across periods.

**D2: Neighborhood composition.** For each evaluation sentence, we use  $h_s$ , the final representation of its subject, as the nearest-neighbor query. For example, for S12 V2 07 at  $t_0$ , the query is the vector at the S12 position. We compare it with the representations of subject occurrences in all other evaluation sentences and retrieve the  $k = 10$  nearest by cosine similarity. For each query, the score is the proportion of neighbors generated from  $N_1$ : if 8 of the 10 neighbors come from  $N_1$ , the score is 0.8. We then average it by trajectory class and period. The reported  $\Delta$  is the score at  $t_9$  minus the score at  $t_0$ ; for example, a decrease from 0.8 to 0.2 gives  $\Delta = -0.6$ , indicating that the neighborhood composition changed from mostly  $N_1$  to mostly  $N_2$ . We also compare the per-period scores with the predefined  $P(N_1 | s, t)$  trajectories.

**D3: Contrastive diagnostic.** D3 asks whether changing only the period changes the masked-verb prediction in the correct direction. Each of the 29 pairs contains the same subject–object frame at an early and a late period. For example, the model receives S12 [MASK] 02 at both  $t_0$  and  $t_9$ . At each period, we sum the predicted probabilities for the  $N_1$  verbs (V1–V4) and for the  $N_2$  verbs (V5–V8). The pair counts as a correct sign flip if the  $N_1$  total is higher at  $t_0$  and the  $N_2$  total is higher at  $t_9$ . A sign-flip rate of 0.3 means that 30% of the pairs meet both criteria.

**D4: Later-period transfer.** D4 trains a separate linear classifier on standard cue representations from  $t_0$ – $t_7$  and evaluates it without retraining on separate sentences from  $t_8$ – $t_9$ . For example, for S12 V6 07 at  $t_8$ , the classifier receives the vector at the S12 position and should identify  $N_2$  as the group that generated the sentence. Accuracy is the proportion of correct  $N_1/N_2$  predictions.

**Auxiliary layer-wise comparison.** We also compute linear centered kernel alignment (CKA) between the representations produced by pairs of models on the same standard cue sentences, at the encoder input and after each encoder layer. Higher CKA values indicate more similar representation patterns [6].

**Memory ablation.** For Memory-Augmented, we also test an oracle version that replaces each representation-derived mean summary with one built from the known generating-group labels. For example, if 80% of a subject’s training sentences at a given period come from  $N_1$ , its oracle summary is  $0.8v_1 + 0.2v_2$ , where the fixed orthogonal vectors  $v_1$  and  $v_2$  represent  $N_1$  and  $N_2$ . The attention

and gate are unchanged. Because the ordinary model does not receive these labels, the oracle serves only as an analysis tool.

*Implementation details.* All models use the same Transformer encoder configuration:  $d = 64$ , 4 attention heads, 2 layers, feed-forward width 128, dropout 0.1. The period encoding projects 32 sinusoidal features of  $t/(T - 1)$  to  $d$  via a two-layer MLP. Training uses AdamW (lr =  $10^{-3}$ , weight decay =  $10^{-2}$ ) with a cosine annealing schedule, 30 epochs, batch size 64. The corpus contains 5,000 SVO sentences; MLM training uses 3,159 sentences from periods  $t_0$ – $t_7$ ; 763 sentences form the standard cue set held out from MLM training and used for D1–D2. The D1 classifier is fitted and evaluated on this set. Its 615 sentences from  $t_0$ – $t_7$  also train the separate D4 classifier. Another 794 sentences from  $t_8$ – $t_9$  are reserved for D4 and excluded from both MLM and classifier training. Separate validation sentences from all ten periods are used to select the version of each model evaluated in the diagnostics. Thus, D4 tests whether a classifier trained on  $t_0$ – $t_7$  transfers to later-period sentences; it is not a fully held-out forecast because model selection uses validation data from  $t_8$ – $t_9$ .

## 6 Results

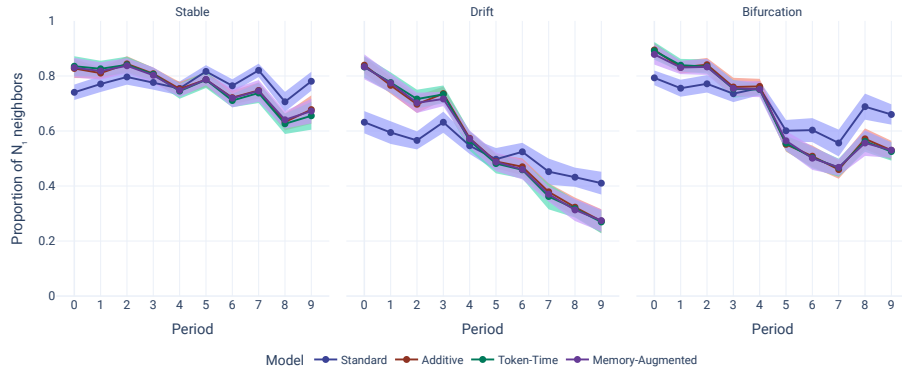
We report means and 95% confidence intervals over 31 training seeds. Period input creates the clearest separation across diagnostics; differences among the time-aware designs are smaller and depend on the evaluation setting. For drifting subjects, all three time-aware models change the proportion of  $N_1$  neighbors by  $\Delta \approx -0.56$ , more than twice Standard’s  $-0.221$ .

### 6.1 Changes in Neighborhood Composition

Figure 3 and Table 1 show how the proportion of  $N_1$  neighbors changes across periods. For drifting subjects, Standard’s mean proportion decreases by 0.221, with a 95% CI of  $[-0.240, -0.203]$ . Token-Time decreases by 0.563, more than twice as much. Additive produces a similar decrease of 0.567.

Stable subjects also show more  $N_2$  neighbors, with  $\Delta$  between  $-0.150$  and  $-0.180$ . This residual change is about one-third of that for drifting subjects ( $\Delta \approx -0.56$ ): the diagnostic distinguishes the trajectory classes, although stable subjects are not unchanged. Bifurcating subjects fall between these cases: the time-aware models reach values near 0.53 at  $t_9$ , close to their predefined 50–50 mixture.

We used a paired equivalence test to determine whether the Additive and Token-Time changes in neighborhood composition are sufficiently close, rather than merely failing to find a difference. We treated absolute differences below 0.05 as equivalent. The mean paired difference (Additive minus Token-Time) is  $-0.003$ , with a 90% CI of  $[-0.026, +0.019]$ ; the TOST supports equivalence ( $p_{\text{TOST}} < 0.001$ ). Equivalence also holds with the narrower margin of 0.034 ( $p_{\text{TOST}} = 0.014$ ).



**Fig. 3.** Proportion of  $N_1$  neighbors. Time-aware models produce stronger  $N_1$ -to- $N_2$  changes for drifting subjects than Standard and similar trajectories to one another. Stable subjects change less; bifurcating subjects end between  $N_1$  and  $N_2$ .

**Table 1.**  $N_1$ -neighbor proportions at  $t_0$  and  $t_9$  (31-seed means;  $\Delta = t_9 - t_0$ ; 95% CI half-widths in brackets).

| Class   | Model      | $t_0$ | $t_9$ | $\Delta$ [CI $_{\pm}$ ] |
|---------|------------|-------|-------|-------------------------|
| Stable  | Standard   | 0.741 | 0.781 | +0.040 [ $\pm 0.017$ ]  |
|         | Additive   | 0.827 | 0.678 | -0.150 [ $\pm 0.027$ ]  |
|         | Token-Time | 0.835 | 0.655 | -0.180 [ $\pm 0.024$ ]  |
|         | Mem-Aug.   | 0.830 | 0.672 | -0.158 [ $\pm 0.021$ ]  |
| Drift   | Standard   | 0.632 | 0.410 | -0.221 [ $\pm 0.018$ ]  |
|         | Additive   | 0.839 | 0.272 | -0.567 [ $\pm 0.019$ ]  |
|         | Token-Time | 0.833 | 0.269 | -0.563 [ $\pm 0.020$ ]  |
|         | Mem-Aug.   | 0.833 | 0.274 | -0.559 [ $\pm 0.022$ ]  |
| Bifurc. | Standard   | 0.793 | 0.660 | -0.133 [ $\pm 0.018$ ]  |
|         | Additive   | 0.895 | 0.530 | -0.365 [ $\pm 0.017$ ]  |
|         | Token-Time | 0.891 | 0.525 | -0.366 [ $\pm 0.016$ ]  |
|         | Mem-Aug.   | 0.878 | 0.530 | -0.348 [ $\pm 0.015$ ]  |

## 6.2 Classifier and Contrastive Diagnostics

The linear classifier reveals a difference that neighborhood composition does not. In the standard cue condition, all time-aware models score above Standard (0.875): Additive reaches 0.892, while Token-Time and Memory-Augmented both reach 0.881.

The ordering changes when verbs and objects provide no reliable contextual cue: Standard reaches 0.565, Additive 0.579, Token-Time 0.595, and Memory-Augmented 0.598. Token-Time is 1.6 percentage points above Additive, and their 95% confidence intervals do not overlap.

Classifier accuracy cannot establish temporal traceability by itself: it tests whether the generating group can be predicted, not whether the composition of a word occurrence’s neighborhood follows changes in use across periods.

**Table 2.** Main results (31-seed means; 95% CIs in brackets). Nbr.  $\Delta$ :  $t_9 - t_0$  proportion of  $N_1$  neighbors for drift; Flip: D3 sign-flip rate; Ambig./Later: classifier accuracy on  $h_s$ .

| Model      | Nbr. $\Delta$        | Flip                | Ambig.              | Later               |
|------------|----------------------|---------------------|---------------------|---------------------|
| Standard   | -0.221 $[\pm 0.018]$ | 0.000               | 0.565 $[\pm 0.004]$ | 0.717 $[\pm 0.004]$ |
| Additive   | -0.567 $[\pm 0.019]$ | 0.602 $[\pm 0.053]$ | 0.579 $[\pm 0.005]$ | 0.746 $[\pm 0.007]$ |
| Token-Time | -0.563 $[\pm 0.020]$ | 0.624 $[\pm 0.048]$ | 0.595 $[\pm 0.006]$ | 0.748 $[\pm 0.005]$ |
| Mem-Aug.   | -0.559 $[\pm 0.022]$ | 0.587 $[\pm 0.051]$ | 0.598 $[\pm 0.008]$ | 0.739 $[\pm 0.008]$ |

All time-aware models produce sign flips at rates from 0.587 to 0.624; Standard scores 0.000 without period input. Their confidence intervals overlap, so D3 does not distinguish among the temporal variants.

In later-period transfer, Standard reaches 0.717, versus 0.746 for Additive, 0.748 for Token-Time, and 0.739 for Memory-Augmented (Table 2). Additive and Token-Time thus behave similarly here and in neighborhood composition; Token-Time’s advantage appears only when contextual words are uninformative.

### 6.3 Memory Ablations

Memory-Augmented remains above Standard on later-period transfer (0.739 vs. 0.717) but not Token-Time (0.748); the paired difference is  $-0.009 [\pm 0.010]$ . The oracle shows that the mechanism can use informative history: label-derived summaries raise later-period accuracy to 0.762  $[\pm 0.006]$ , 0.013 above Token-Time  $[\pm 0.009]$ . Its sign-flip rate is 0.694  $[\pm 0.118]$ , 0.070 above Token-Time, although this estimate has a wide confidence interval.

**Table 3.** Memory ablations (31-seed means; 95% CIs in brackets). Later: D4 classifier accuracy at  $t_8 - t_9$  after fitting at  $t_0 - t_7$ ; Flip: D3 sign-flip rate.

| Model             | Later               | Flip                | $\Delta$ later vs. Token-Time |
|-------------------|---------------------|---------------------|-------------------------------|
| Token-Time (ref.) | 0.748 $[\pm 0.005]$ | 0.624 $[\pm 0.048]$ | –                             |
| Mem-Aug. mean     | 0.739 $[\pm 0.008]$ | 0.587 $[\pm 0.051]$ | $-0.009 [\pm 0.010]$          |
| Mem-Aug. oracle   | 0.762 $[\pm 0.006]$ | 0.694 $[\pm 0.118]$ | $+0.013 [\pm 0.009]$          |

To test whether a single mean summary discards a possible two-group structure, we replaced it with two  $k$ -means cluster centers per subject and period. The rest of the memory architecture remains unchanged. Across 15 matched seeds, this alternative does not improve later-period accuracy or the sign-flip rate over either mean pooling or Token-Time; all four confidence intervals for the differences include zero.

## 7 Discussion and Conclusion

Temporal conditioning more than doubles the change in neighborhood composition for drifting subjects relative to Standard, and this change is about three

times the residual change observed for stable subjects. Under reliable contextual cues, direct addition is as effective as a learned token–period interaction, making Additive the more parsimonious design in this setting.

Additive and Token-Time are equivalent when verbs and objects provide reliable contextual cues (TOST,  $\delta = 0.05$ ,  $p_{\text{TOST}} < 0.001$ ). When those words provide no reliable cue, the classifier gives Token-Time a small advantage over Additive (0.595 vs. 0.579). The layer-wise CKA comparison tells a complementary story: Additive–Token-Time similarity rises from 0.726 at the input to 0.887 after the encoder. Their overall representation patterns become more similar with depth, while the ambiguous condition still reveals a classifier difference.

If ambiguity systematically favored Token-Time, its advantage should grow as contextual cues become less reliable. The 15-seed noise sweep does not show this pattern. For D2, the slope is  $+0.093$  (95% CI  $[-0.279, +0.464]$ ,  $d_z = 0.14$ ,  $p = 0.601$ ); for classifier accuracy, it is  $-0.019$  (95% CI  $[-0.105, +0.067]$ ,  $d_z = -0.12$ ,  $p = 0.644$ ). The ambiguous cue result is therefore evidence for a specific condition, not general Token-Time superiority.

The memory ablations further narrow the design space. The oracle shows that the existing attention and gate can exploit informative history, while neither one representation-derived mean nor two unsupervised  $k$ -means cluster centers surpass Token-Time. This suggests that the bottleneck lies in the information recovered by these unsupervised summaries, rather than in mean pooling alone.

These results position the suite as a controlled architectural audit before evaluation on natural LSCD data. RuShiftEval, for example, follows the same words across three periods and includes change followed by stability as well as continuous change [7]. With annotated targets, stable words can serve as negative controls, while gradual change and the gain or coexistence of senses provide approximate counterparts to our drifting and bifurcating trajectories.

A natural-corpus study can test whether Additive and Token-Time remain equivalent in neighborhood composition when local contexts identify the current sense and whether Token-Time gains an advantage for more ambiguous usages. Such a study must adapt the diagnostics to imperfectly annotated trajectories and account for open-ended polysemy, metaphor, register variation, frequency shifts, and genre differences, none of which the controlled SVO corpus reproduces.

In this controlled setting, direct addition matches more complex temporal conditioning in average changes to neighborhood composition. Token-Time’s localized advantage under ambiguity identifies where a learned interaction may matter, while the memory ablations show that useful history can help when its summary preserves the relevant distinction. More broadly, period-identifier or generating-group recovery is not enough: temporal traceability requires contextual representations conditioned on the requested period whose neighborhoods can be inspected across time and whose changes reflect changes in how the word is used.

## References

1. Conneau, A., Kruszewski, G., Lample, G., Barrault, L., Baroni, M.: What you can cram into a single  $\$ \&! \#^*$  vector: Probing sentence embeddings for linguistic properties. In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 2126–2136 (2018)
2. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL). pp. 4171–4186 (2019)
3. Dhingra, B., Cole, J.R., Eisenschlos, J.M., Gillick, D., Eisenstein, J., Cohen, W.W.: Time-aware language models as temporal knowledge bases. Transactions of the Association for Computational Linguistics (TACL) **10**, 257–273 (2022)
4. Giulianelli, M., Del Tredici, M., Fernández, R.: Analysing lexical semantic change with contextualized word representations. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 3960–3973 (2020)
5. Hamilton, W.L., Leskovec, J., Jurafsky, D.: Diachronic word embeddings reveal statistical laws of semantic change. In: Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 1489–1501 (2016)
6. Kornblith, S., Norouzi, M., Lee, H., Hinton, G.: Similarity of neural network representations revisited. In: Proceedings of the 36th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 97, pp. 3519–3529. PMLR (2019)
7. Kutuzov, A., Pivovarova, L.: Three-part diachronic semantic change dataset for Russian. In: Proceedings of the 2nd International Workshop on Computational Approaches to Historical Language Change. pp. 7–13. Association for Computational Linguistics (2021). <https://doi.org/10.18653/v1/2021.lchange-1.2>
8. Loureiro, D., Barbieri, F., Neves, L., Anke, L.E., Camacho-Collados, J.: TimeLMs: Diachronic language models from Twitter. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL): System Demonstrations. pp. 251–260 (2022)
9. Rosin, G., Radinsky, K.: Time masking for temporal language models. In: Proceedings of the 15th ACM International Conference on Web Search and Data Mining (WSDM). pp. 833–841 (2022)
10. Schlechtweg, D., McGillivray, B., Hengchen, S., Dubossarsky, H., Tahmasebi, N.: SemEval-2020 task 1: Unsupervised lexical semantic change detection. In: Proceedings of the 14th International Workshop on Semantic Evaluation (SemEval). pp. 1–23 (2020)
11. Tahmasebi, N., Borin, L., Jatowt, A.: Survey of computational approaches to lexical semantic change. Computational Approaches to Semantic Change pp. 1–91 (2021)
12. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 30, pp. 5998–6008 (2017)
13. Zamora-Reina, F.D., Bravo-Marquez, F., Schlechtweg, D.: LSCDiscovery: A shared task on semantic change discovery and detection in Spanish. In: Proceedings of the 3rd Workshop on Computational Approaches to Historical Language Change (LChange). pp. 149–164 (2022)